Research Project – Log Book

IT22627582 – Pathiraja H.P.M.H.D.S. – Rework Counter – 22.03.2026 to 18.10.2026

Supervisor	Mrs. Narmada Gamage
Co-Supervisor	Mrs. Budhdhima Aththanayaka
Project-ID	R26-IT-054 – AI Powered Vision System for Realtime Garment Tracking in Apparel Manufacturing
Student-ID	IT22627582
Name	Pathiraja H.P.M.H.D.S.
Individual Component	Cloth Piece Detecting and Counting Reworks and Good Works.

Summary of the Individual Component

This component introduces an offline-first, real-time computer vision system designed for factory floor automation. It processes live workstation video feeds to detect, track, and classify garment pieces as "Good" or "Rework" based on their movement through predefined digital zones. The architecture features a Python/FastAPI sidecar processing YOLO inference and ByteTrack object tracking, seamlessly proxying the video stream through custom Electron protocols (garmentrework://) to a React dashboard. Significant engineering focus was applied to mitigate hardware constraints on standard factory machines, including dynamic resolution downscaling, frame rate throttling, and preventing SQLite/Event-loop deadlocks, ensuring continuous UI responsiveness and accurate tracking during heavy CPU loads.

March 2026

Date	RP Work
Mar 22, 2026	Started the project after the proposal presentation.Listed the hardware limits for factory computers at MAS Capital.
Mar 24, 2026	Created a GitHub repository for the project to save code.Set up the basic folder structure for the frontend and backend files.
Mar 26, 2026	Read research papers on fast object tracking methods.Set up Python tools and GPU settings for AI model training.
Mar 28, 2026	Tested the laptop webcam with basic OpenCV code.Checked how to read and display live video frames using Python.
Mar 30, 2026	Talked with MAS Capital to plan the factory visit.Prepared cameras to record videos in different factory lighting conditions.

April 2026

Date	RP Work
Apr 01, 2026	Created simple wireframe sketches for the React dashboard.Planned where to put the video screen, buttons, and settings sliders.
Apr 03, 2026	Visited the garment factory to collect real video data.Recorded live videos of the quality check (QC) process on the production line.
Apr 06, 2026	Watched the factory videos to find the best camera angles.Removed blurry or unusable video clips from the dataset.
Apr 08, 2026	Saved clear images from the factory videos.Drew boxes around the garments in the images to train the AI.
Apr 10, 2026	Learned how to use the Ultralytics YOLOv8 AI library.Wrote a small script to test the YOLO library on a single factory image.
Apr 15, 2026	Trained the YOLO AI model using the custom factory images.Tested the model and found issues when garments covered each other on the table.
Apr 18, 2026	Tested the trained model on different moving factory videos.Took notes on false detections and missing garment IDs during the video.
Apr 22, 2026	Learned about ByteTrack to fix the garment covering issue.Added ByteTrack to the code to track garments correctly across video frames.

Apr 25, 2026	Combined the YOLO boxes and ByteTrack IDs together.Wrote code to print the tracking ID on the screen for each garment.
Apr 28, 2026	Wrote Python code to find the center point of each garment.Drew top and bottom boundary lines on the video to count "Good" and "Rework" items.

May 2026

Date	RP Work
May 02, 2026	Added logic to check if a garment crossed the center zone.Wrote conditions to stop the system from counting the same item twice.
May 05, 2026	Created an offline database (SQLite) to save inspection results.Changed the simple Python code into a FastAPI web server.
May 08, 2026	Tested the new FastAPI server endpoints using Postman.Fixed a CORS error that blocked the frontend from connecting to the backend.
May 10, 2026	Created system diagrams for the upcoming progress presentation.Checked the accuracy of the tracking model using test videos.
May 12, 2026	Presented the first progress update (Progress Presentation 1) to the panel.Took notes on panel feedback about making the app run smoothly on slow computers.
May 13, 2026	Submitted the mandatory Check List to the university.Planned new tasks to reduce the CPU usage based on panel feedback.
May 17, 2026	Learned how to send images over a web connection.Researched the special format needed for continuous live video streaming.
May 20, 2026	Wrote code to send live video frames from the Python server to the frontend.Set up the correct video formats to make the stream play smoothly.
May 24, 2026	Installed the Lucide-React library to get nice icons for the dashboard.Created the side navigation bar and top header parts of the UI.

May 27, 2026	Started building the main user interface using React and Tailwind CSS.Designed the main screen for camera selection and live video viewing.
May 30, 2026	Built the "Order Analytics" section in the user interface.Created the input fields for the user to type the "Avg Rework Fix Time".

June 2026

Date	RP Work
Jun 02, 2026	Added JavaScript logic to find all connected cameras on the computer.Created a dropdown list for the user to select the correct camera.
Jun 04, 2026	Set up the Electron framework to make a standalone desktop app.Wrote code to automatically find a free network port for the Python server.
Jun 08, 2026	Wrote the configuration file loader for the Electron app.Set up the computer paths so Electron knows where the Python files are.
Jun 11, 2026	Made the Electron app start the Python server silently in the background.Fixed bugs related to starting and stopping the hidden background server.
Jun 15, 2026	Created a health-check system to see if Python is ready.Made the React app show a loading screen until the Python server finishes loading the AI.
Jun 18, 2026	Connected the React buttons to the Python backend to start tracking.Added settings to change AI confidence levels and zone sizes in real-time.
Jun 21, 2026	Wrote code to generate a unique "Session ID" every time the user clicks "Start".Sent this ID to the Python backend to save data under the correct session.
Jun 25, 2026	Created a special video link format (garmentrework://) for security.Set up the app to safely route this link to the hidden Python server.
Jun 28, 2026	Styled the Good, Rework, and Pass Rate cards using Tailwind CSS.Added color-coded borders and large text sizes for the numbers.

July 2026

Date	RP Work
Jul 02, 2026	Tested the whole system together for the first time.Found a bug where the video feed was blocked by Electron security rules.
Jul 06, 2026	Read Electron documentation about web app security.Learned how the Content Security Policy (CSP) affects images and video streams.
Jul 09, 2026	Fixed the video block by updating the Content Security Policy (CSP).Confirmed that the live video now shows up correctly in the React app.
Jul 13, 2026	Added the "LIVE INSPECTION RUNNING" green badge to the UI.Made the badge pulse when the video is active and playing.
Jul 16, 2026	Discovered a big issue where the video stopped playing after a few seconds.Found that the AI was using 100% of the CPU and crashing the connection.
Jul 20, 2026	Tested different video sizes to see which one uses less CPU power.Chose 640x480 as the best balance for speed and video quality.
Jul 24, 2026	Made the video frame smaller before sending it to the AI.Successfully reduced the CPU usage by 60% without losing accuracy.
Jul 28, 2026	Removed the async keyword from the Python video function.Verified that this change helped the server run other tasks without freezing.

August 2026

Date	RP Work
Aug 02, 2026	Found that reading and writing to the database at the same time caused system freezes.Changed the code to read counts directly from memory instead of the database.
Aug 06, 2026	Wrote the logic to save the last 50 events in a computer memory list.Sent this list to the React UI to show a live log without touching the database.
Aug 10, 2026	Fixed an issue where the Electron app cut the video connection exactly at 10 seconds.Removed the 10-second timeout limit from the video fetching code.
Aug 14, 2026	Noticed the video crashing completely on some tests.Used Developer Tools to find the net::ERR_FAILED network error.
Aug 18, 2026	Found a tricky bug where the computer deleted the video stream automatically (Garbage Collection).Saved the video stream in a global variable to stop the computer from deleting it.
Aug 22, 2026	Left the application running for 2 hours to test stability.Confirmed the video did not crash or freeze over the long test period.
Aug 26, 2026	Tested the user interface to make sure it runs fast without freezing.Got the system ready for the second progress presentation.
Aug 29, 2026	Created PowerPoint slides for the upcoming presentation.Practiced explaining how the CPU and memory bugs were fixed.

September 2026

Date	RP Work
Sep 01, 2026	Showed the fully working, bug-free app to the panel (Progress Presentation 2).Demonstrated the live video tracking working smoothly without high CPU usage.
Sep 05, 2026	Received feedback to support pre-recorded videos better for demonstrations.Started working on logic to make video files play in a loop.
Sep 08, 2026	Fixed a bug where the app crashed at the very end of a video file.Added code to make video files loop continuously from the beginning.
Sep 12, 2026	Fixed a React warning about the chart size not being set correctly.Ensured the UI layout adjusts perfectly on different screen sizes.
Sep 15, 2026	Improved the React code to stop unnecessary background tasks.Cleaned up memory usage when the user switches to different screens.
Sep 19, 2026	Tested the whole application on a low-end laptop to simulate factory PCs.Checked if the AI still detects garments correctly at lower frame rates.
Sep 22, 2026	Tested building the final Windows desktop application (.exe).Fixed issues where the app could not find the Python files after packaging.
Sep 27, 2026	Wrote a Python script to automatically install all required libraries.Tested the setup script to ensure the final build works on a new computer.

October 2026

Date	RP Work
Oct 02, 2026	Added error messages to tell the user if the camera disconnects.Made sure the app does not crash completely if the Python server stops working.
Oct 05, 2026	Wrote the introduction and literature review chapters for the thesis.Formatted the thesis document according to the university guidelines.
Oct 08, 2026	Wrote the methodology chapter for the final thesis.Drew system architecture diagrams to include in the thesis report.
Oct 11, 2026	Submitted the Draft Thesis to the university system.Finished and submitted the Research Project Website.
Oct 14, 2026	Checked all final requirements and system features one last time.Submitted the Final Check List to the university.
Oct 18, 2026	Compared the final fast AI model with the old slow one to write the results.Prepared this logbook for the final logbook submission and viva presentation.